Nuclear Instruments and Methods in Physics Research A 607 (2009) 458–462

Contents lists available at ScienceDirect

Nuclear Instruments and Methods in Physics Research A

journal homepage: www.elsevier.com/locate/nima

Optimized readout configuration for PIXE spectrometers based on Silicon Drift Detectors: Architecture and performance

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ARTICLE INFO

Article history:
Received 13 February 2009
Received in revised form
18 May 2009
Accepted 25 May 2009
Available online 11 June 2009

Keywords:
Silicon Drift Detectors
Front-end electronics
Charge preamplifier
PIXE
Backscattered protons

ABSTRACT

An optimized readout configuration based on a charge preamplifier with pulsed-reset has been designed for Silicon Drift Detectors (SDDs) to be used in Particle Induced X-ray Emission (PIXE) measurements. The customized readout electronics is able to manage the large pulses originated by the protons backscattered from the target material that would otherwise cause significant degradation of X-ray spectra and marked increase in dead time. In this way, the excellent performance of SDDs can be exploited in high-quality proton-induced spectroscopy of low- and medium-energy X-rays. This paper describes the designed readout architecture and the performance characterization carried out in a PIXE setup with MeV proton beams.

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1. Introduction

It is widely acknowledged that Silicon Drift Detectors (SDDs)(introduced in 1983 by Gatti and Rehak [1]) offer the opportunity of performing high-resolution and high-count-rate X-ray spectroscopy close to room temperature [2–5], with superior performance with respect to the conventional Si(Li) detectors and pn diodes. The better performance of SDDs can be successfully exploited, in particular for the detection of low–medium energy X-rays, for which the electronic noise contribution to the overall energy resolution is relevant (at higher X-ray energy, in fact, the statistic contribution is dominant) and the reduced available thickness (450 μm) is not a limit.

When such detectors are applied in Particle Induced X-ray Emission (PIXE) measurements, however, also particles backscattered from the target material reach the detector chip. Although the rate of backscattered particles is generally one or two orders of magnitude lower than the X-ray rate—backscattering cross-sections are lower than PIXE cross-sections—the energy released all at once in the detector can be up to a few MeV, i.e., even one–two orders of magnitude higher than the energy of detected X-rays [6]. Without optimized front-end electronics, this can lead to several negative effects: circuit saturation, marked increase in the system dead time and degradation of the energy resolution due to signal-dependent noise. In practice, X-ray spectra can be significantly spoiled already at very low proton rates (i.e. low beam current and low X-ray rate). A further effect is the permanent performance degradation of the chip owing to proton implantation in the silicon active volume, which appears however only after a relatively high dose of backscattered protons [7].

When low-energy X-rays (from 1 keV upward) are to be detected, absorber foils cannot be used to prevent protons from reaching the detector. A common choice to avoid the negative effects caused by backscattered particles is the use of magnetic deflectors in front of the X-ray detector [8]. This usually implies, however, also the reduction of the detector solid angle, thus decreasing the X-ray detection efficiency, and may make system alignment complicate.

With the motivation of fully exploiting the excellent performance of Silicon Drift Detectors in PIXE, we studied alternative methods to get rid of the negative effects of backscattered protons, following two different directions. On the one hand, we have developed an SDD-based spectrometer coupled to a polycapillary lens, which ensures that X-rays are efficiently transported to the detector while the proton rate is significantly reduced (see [9] for details). On the other hand, we have designed customized readout electronics, capable of dealing with the large signals caused by protons without spoiling the SDD performance. This solution obviously does not avoid the chip damage (which is

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doi:10.1016/j.nima.2009.05.165

anyway a long-period problem for the typical experimental conditions used in PIXE analyses). However, as an additional benefit, it may improve performance in the presence of large signals caused by a gamma-ray background.

This work illustrates the optimized readout configuration for an SDD-based PIXE spectrometer with a pulsed-reset charge preamplifier. After the description of the readout architecture, the measured performance of the system is reported, as experimentally characterised in PIXE measurements with proton beams of few MeV at the LABEC accelerator of INFN Florence.

2. Architecture of the optimized readout based on a pulsed-reset charge-amplifier configuration

The considered detector is a Silicon Drift Detector with an active area of 10 mm^2 and thickness of about 450 μm in polysilicon technology provided by PNSensor GmbH. Such detector features symmetric round shape and concentric drift rings with the anode ring in the center. The anode area hosts a circular on-chip JFET integrated on the same detector chip with all the constraints of a detector-grade technology [10]. Different readout strategies, summarized in [11–13], can be implemented with the on-chip FET. We chose a charge preamplifier configuration with pulsed-reset optimized to work in the presence of large signals as those caused by proton interaction. Fig. 1 shows the simplified schematic of the designed configuration. The on-chip FET is biased in source-follower configuration and followed by an AC-coupled trans-resistance amplifier. In this way it is possible to decouple detector and on-chip JFET biasing from the voltages required for the proper operation of the preamplifier. At very low frequencies, the system operates nearly in an open-loop configuration and the DC level of the anode is set by the reset mechanism that will be discussed later. The p+ electrode close to the anode, whose parasitic capacitance is used as feedback capacitor (C_{a-ig}), is biased through a large-value resistor (R_b in Fig. 1) so that the preamplifier output can drive the feedback capacitor. The resistance of resistor R_b and the capacitance of capacitor C_c are chosen so that they do not influence the circuit behavior.

In the frequency range that ensures a high voltage gain between the on-chip JFET gate and the preamplifier output, the whole preamplifier operates in closed-loop configuration and the gate of the on-chip JFET acts as virtual ground. Therefore the signal current is integrated over the feedback capacitance C_{a-ig}. The capacitor C_p represents the stray capacitance caused by the connection from the detector chip to the preamplifier board. A sufficiently large value of the capacitor C_{in} results in a pole at sufficiently low frequency. In this way, both the leakage current

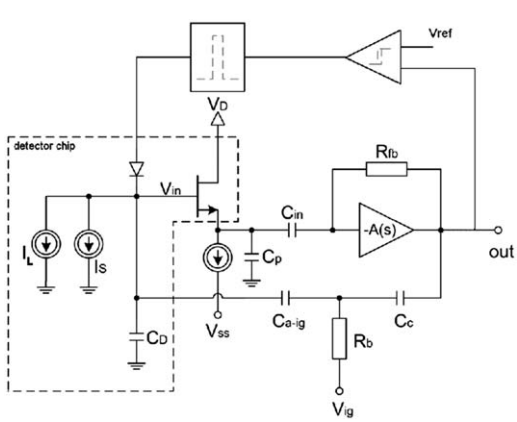

Fig. 1. Circuit schematic of the readout configuration.

and the signal current are integrated on the feedback capacitor C_{a-ig}, providing the following ideal transfer function:

$$T_{id}(s) = \left.\frac{V_{out}(s)}{I_{in}(s)}\right|_{id} = -\frac{1}{sC_{a-ig}} \tag{1}$$

By computing the loop gain we can derive the actual transfer function of the circuit:

$$T(s) = \frac{T_{id}(s)}{1-\frac{1}{G_{loop}(s)}} = -\frac{C_{a-ig}}{C_D + C_{a-ig}}\frac{R_{fb}C_{in}}{C_{a-ig}}\frac{1}{1+s\left(\frac{C_{a-ig}}{C_D+C_{a-ig}}R_{fb}C_{in}+\frac{C_{in}}{g_{mFET}}\right)} \tag{2}$$

Therefore the circuit behaves as an ideal integrator only in the frequency range larger than

$$f_t = \frac{1}{2\pi(C_{a-ig}/(C_D + C_{a-ig})R_{fb}C_{in} + C_{in}/g_{mFET})} \tag{3}$$

that is of the order of few Hz for typical values of the components.

A rate-adaptive reset provides the pulsing to the reset diode. The output voltage of the preamplifier is compared with a preset value and the comparator output triggers a multivibrator stage that provides the pulse to be applied to the reset diode.

3. Experimental setup and conditions for the performance characterization

The detector (naked chip) is mounted on a ceramic carrier coupled with a copper heat sink, housed in a vacuum-tight case. The readout electronics board is located as close as possible to the detector itself on a separated board connected via a flat cable. A Peltier stage is used to cool the copper heat sink and provides a detector working temperature of −30 °C. The entrance window is a 8-μm-thick Be layer that allows efficient medium-light element X-ray detection (window transmittance is still higher than 40% at 1 keV). No collimator is placed in front of the detector active area.

The spectrometer with a pulsed-reset charge amplifier configuration was characterized at one of the external beamlines of the 3 MV Tandem accelerator of the LABEC laboratory (INFN, Florence).

The spectrometer was placed at an angle of 135° with respect to the beam direction, in a geometry that is typical for PIXE measurements aimed at the detection of low- and medium-energy X-rays. In particular, helium was continuously flowing in front of the detector entrance window in order to enhance the transmission of low-energy X-rays. The overall detector-to-target distance was about 60 mm (about 20 mm in vacuum and 40 mm in atmosphere, three-fourths of which fully saturated with helium).

We used external proton beams with 3, 4 and 5 MeV energy (nominal values prior to extraction). After the 8 μm Upilex beam exit window and about 1 cm beam path in atmosphere, the actual energy values at the target surface were calculated to be 2.85, 3.9 and 4.9 MeV. Beam spots of about 500-μm diameter were defined by collimation (before the exit window).

The performance of the system was evaluated mainly using a thick zinc standard as target, with characteristic X-rays of about 1 keV (L lines), 8.6 keV (Kα line) and 9.6 keV (Kβ line). Since the target is thick, the energy spectrum of the backscattered protons ranges from zero to a maximum value, determined by the incident beam energy and the experimental geometry and conditions. The maximum energy of backscattered protons reaching the detection chip in the described geometry is about 2.3, 3.4 and 4.4 MeV with 3, 4 and 5 MeV incident beam energy, respectively. For each energy, measurements were carried out at various levels of beam current intensity in order to probe the system response (detector+preamplifier) at different rates of X-rays and different

rates of backscattered protons. The backscattered-proton rate was monitored on-line by means of a customized logic circuit that provides a digital pulse for every detected signal over a given threshold and then by a digital counter (Agilent 53131A Universal Frequency Counter).

4. Characterization of the circuit functionality

The first characterization aimed at testing the system response in terms of shape of the output waveform for both X-ray and proton interactions. Fig. 2 shows the system output waveform when the proton beam is deflected from the beamline and therefore no protons impinge on the sample. The slope caused by the integration of the leakage current is clearly visible. With the detector cooled at $-30\,^{\circ}$C and the threshold set at 1.6 V, the reset period is 26.5 ms. The inset shows a detail of the preamplifier output waveform during reset. The preamplifier is ready for a novel pulse acquisition after less than 10 μs in the case of a normal reset. We have also qualified the spectroscopy performance of the system by means of irradiation with a ^{55}Fe radioactive source. The achieved energy resolution (with no changes in repeated measurements) is of the order of 140 eV FWHM at the Mn Kα line at 1 μs shaping time (7th order pseudo-Gaussian shaper).

We have investigated the reset performance in the case of protons impinging on the detector at different energies. Fig. 3 shows the typical preamplifier output waveform during PIXE acquisition with a 3 MeV proton beam. Both protons and X-rays impinge on the detector. In the case of a proton interaction, a reset pulse is immediately applied to the reset diode in order to remove the deposited charge from the feedback capacitance and therefore restore the DC level of the anode potential.

Fig. 4 shows the preamplifier output waveforms for different types of radiations interacting with the detector (X-rays and protons) when the incident beam energy was 5 MeV (nominal value). In particular, Fig. 4a shows the voltage step caused at the preamplifier output by an X-ray (in this case the energy corresponds to a Zn Kβ X-ray). Fig. 4b–d refer to the detection of backscattered protons impinging on the detector with increasing energies. In the case of Fig. 4b, the energy released by the proton is low enough and can be removed by a single reset pulse. The time delay (about 1.5 μs) between the voltage step caused by the integration of the charge released by the proton and the reset pulse is mainly due to the saturation of the inner stages of the preamplifier, while the shape of the reset pulse is comparable with the one of the so-called normal reset. Since the main concern is to avoid dead time and preamplifier saturation in the case of energetic protons impinging on the detector, it is critical to properly choose the timing properties of the reset pulse in terms of starting time and duration. In any case the total dead time is well below 10 μs. When the amount of charge released in the detector and therefore integrated on the feedback capacitor is larger, as shown in Fig. 4c and d, several pulses are needed to fully remove the charge released by the proton from the anode and restore the DC anode voltage. The time delay between the voltage step caused by the integration of the charge released by the proton and the reset pulse stays nearly constant between 1.2 and 1.5 μs. Each reset pulse has a fixed duration (5 μs in the present case) and during this time the forward bias of the reset diode is not able to remove the total amount of charge deposited on the feedback capacitance; therefore when the diode is set back to reverse bias, there is still a remaining part of the signal charge on the feedback capacitance that causes the preamplifier output voltage to be again over-threshold. The higher the proton energy, the larger the deposited charge and the larger the number of reset pulses needed. As shown in Fig. 5, it happens sometimes that after

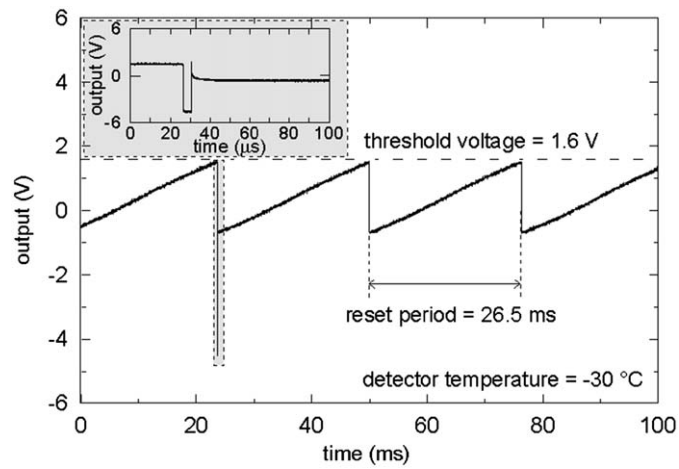

Fig. 2. Preamplifier output waveform when no photons impinge on the detector. The inset shows a detail of the reset pulse. The slope caused by the integration of the leakage current and the reset pulses are clearly visible.

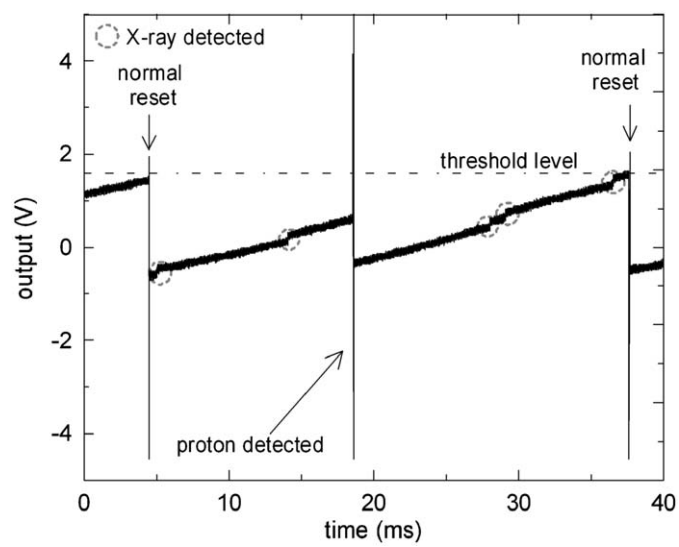

Fig. 3. Typical preamplifier output waveform during PIXE acquisition. Both proton and X-rays impinge on the detector. In the case of proton interaction a reset pulse is immediately applied to the reset diode. The incident beam energy was set to 3 MeV (nominal value).

the last reset pulse that the circuit is able to provide, there is still a fraction of the signal charge deposited on the feedback capacitor, but such amount is not enough to cause an output voltage able to trigger a novel reset. Therefore the final DC output level is higher than in the case of a normal reset. It has to be pointed out that the large and sudden variation of the preamplifier output that occurs after proton detection before the reset pulse is applied causes a variation of the bias point of the p+ electrode close to the anode, whose parasitic capacitance is used as a feedback capacitor that turns in a variation of the feedback capacitance (C_{a-ig}). Hence it is critical that the reset pulse is immediately applied. Even if more energetic protons cause an increase of the dead time, for all the experimented situations the dead time stays below 40 μs.

The preamplifier output feeds a shaping amplifier (Silena 7611L in our setup) that processes the preamplifier output waveform and shapes every positive voltage step. In our setup, the amplitude of the shaper output pulse in the case of proton detection is out of the ADC conversion range and in any case the total dead time stays below 40 μs. If needed, an inhibit signal can be derived from the preamplifier to prevent acquisition of these signals.

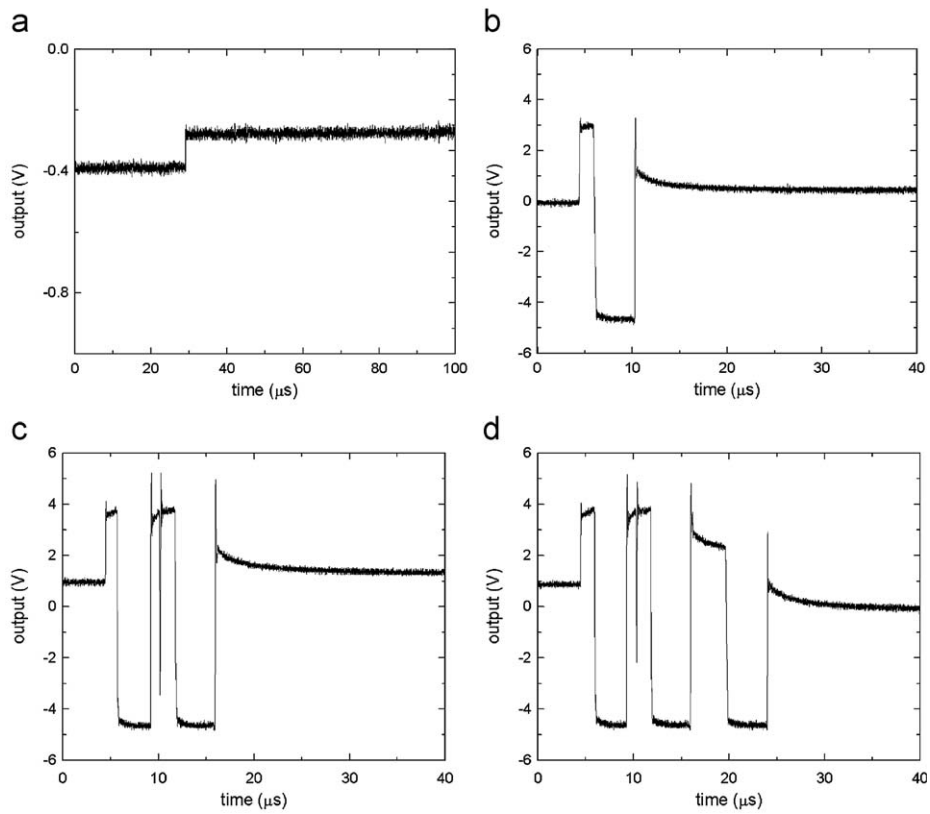

Fig. 4. Preamplifier output waveforms for an X-ray (the energy corresponds to a Zn Kβ X-ray) (a) and for backscattered protons impinging on the detector with different energy values (b–d). The incident beam energy was set to 5 MeV (nominal value). For lower energy protons that impinge on the detector the dead time is shorter and a lower number of reset pulses is needed for the complete anode reset.

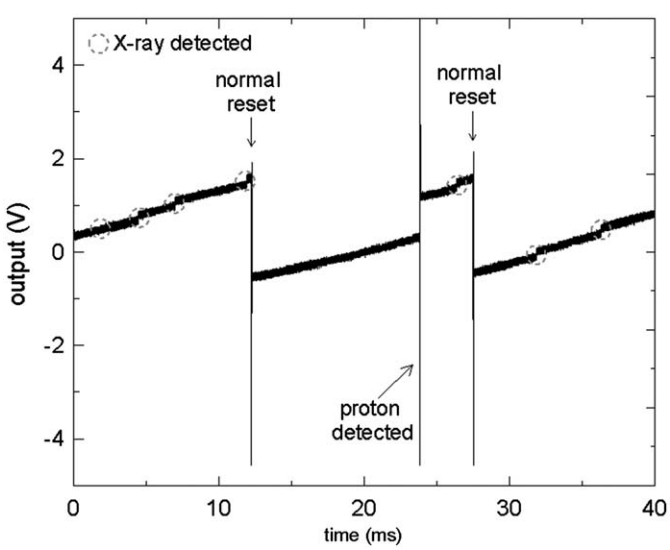

Fig. 5. Preamplifier output waveform during PIXE acquisition. Both proton and X-rays impinge on the detector. The incident beam energy was set to 3 MeV (nominal value). In the case of proton interaction after the last reset pulse a fraction of the signal charge deposited on the feedback capacitor is still present, but such amount is not enough to trigger a novel reset. DC output level is higher than in the case of a normal reset.

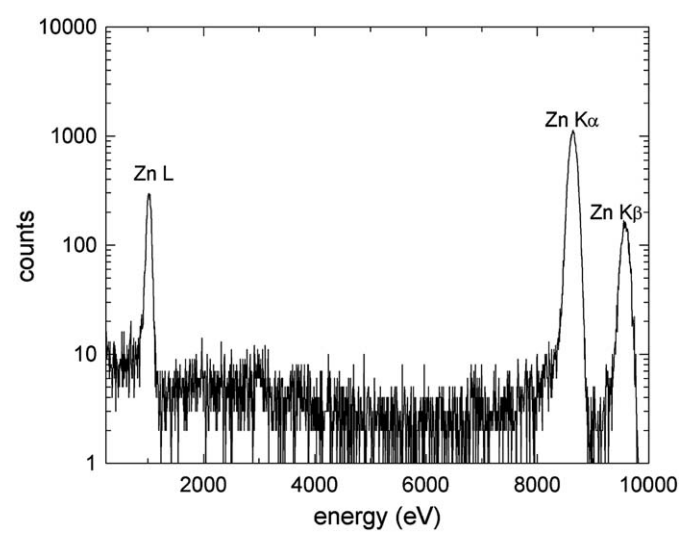

Fig. 6. PIXE spectrum from a thick Zn sample (185 Hz X-ray rate). The incident beam energy was set to 3 MeV (nominal value).

5. Measured performance in PIXE spectroscopy

We have then qualified the spectrometer performance in a true PIXE measurement. Fig. 5 shows the measured energy spectrum with the Zn standard at 185 Hz X-ray rate and 9 Hz proton rate. The beam energy was set to 3 MeV. The measured energy resolution is 161 eV FWHM at the Zn Kα line, consistent with the previously assessed off-beam energy resolution at the Mn Kα line.

By acquiring different spectra at increasing proton rates and at different proton-beam energies, we have verified that the energy resolution does not degrade at increasing count rates even on the low-energy Zn L peak that is in the range where the energy resolution is not Fano-limited. Fig. 6 shows the measured FWHM at the Zn Kα line as a function of the proton rate. The upper axis shows the corresponding total X-ray rate as derived from the

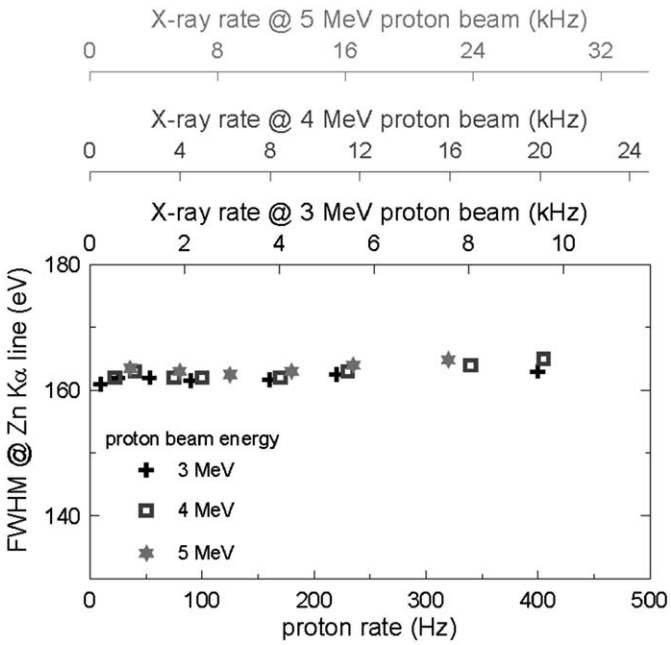

Fig. 7. Measured FWHM at the Zn Kα line as a function of the proton rate on the detector for different energies of the proton beam. The upper axes show the corresponding X-ray photon rate on the detector at the different beam energies.

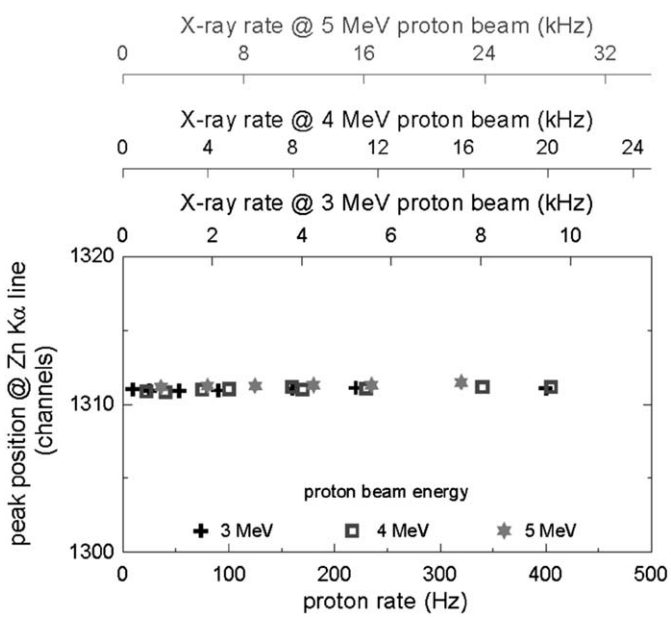

Fig. 8. Measured peak position at the Zn Kα line as a function of the proton rate on the detector for different energies of the proton beam. The upper axes show the corresponding X-ray photon rate on the detector at the different beam energies.

acquired spectra. As it can be noticed, the energy resolution keeps constant for a large variation of the proton rate up to about 400 Hz, despite the large variation of signal-leakage current due to the increased X-ray and proton rate. The maximum worsening of the energy resolution at the Zn Kα line is from 161 eV FWHM when the proton rate is negligible (9 Hz) to 165 eV FWHM with 320 Hz proton rate (5 MeV beam energy) and 19 kHz X-ray rate.

From the same data set, we also evaluated the peak stability. Fig. 7 shows the measured peak position (channel of the Kα line) as a function of the proton rate, and hence of the total X-ray rate. The peak centroid keeps constant (the spread of the measured

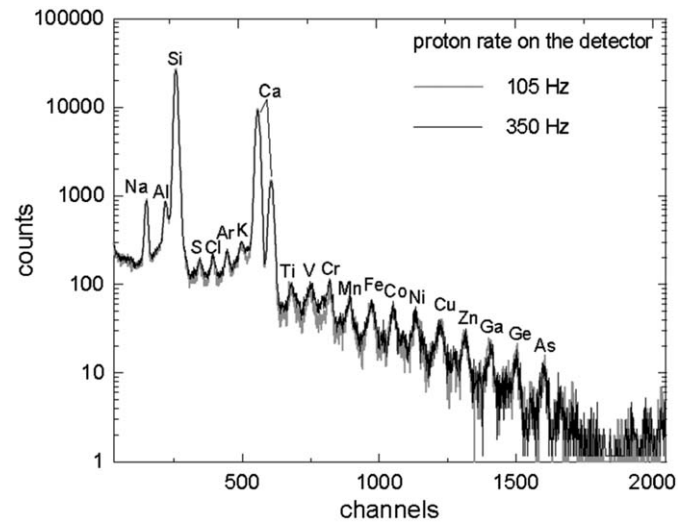

Fig. 9. Comparison of the PIXE spectra of the NIST silicate glass standard SRM 610 (500 ppm trace element glass) acquired with the charge preamplifier with pulsed reset at two different beam currents. The incident beam energy was set to 4 MeV (nominal value).

values is 0.17 channels corresponding to 1.1 eV and to the 0.013% of the peak position) over the investigated range.

For a better performance evaluation, we have measured the PIXE spectrum of a multi-element standard (the NIST silicate glass SRM 610–500 ppm trace element glass) at two different beam currents (therefore two different rates of X-rays and of backscattered protons on the detector). The incident beam energy was set to 4 MeV (nominal value). Fig. 8 shows the measured spectra (normalized to the Si peak) plotted against channels (i.e. without any calibration) to show the excellent peak stability, despite rate variation. The centroid of the Na K line occurs at channel 148.97 at 105 Hz backscattered-proton rate and at 148.99 at 350 Hz (Fig. 9).

Acknowledgements

This work was supported by INFN—Istituto Nazionale di Fisica Nucleare under experiment DANTE (Development of Analytical Nuclear Techniques). The authors gratefully acknowledge P. A. Mandò for many fruitful and stimulating discussions and A. Quattrone for his collaboration in the initial design of the preamplifier.

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